$Var(X) = \sigma^2 = \mathbb{E}[(X - \mu)^2]$], but don't worry about remembering this equation—we will use built-in numpy functions to compute variance. Variance is either denoted *Var*(*X*) or σ^2 (sigma squared) where $\sqrt{\sigma^2} = \sigma$ is the standard deviation, so the variance is the standard deviation squared. The μ in this equation is the standard symbol for mean, which again is $\mu = \mathbb{E}[X]$, where *X* is a *random variable* of interest.

A random variable is just another way of using a probability distribution. A random variable is associated with a probability distribution, and a probability distribution can yield random variables. We can create a random variable, *T*, for tomorrow's temperature. It is a random variable since it is an unknown value but it can only take on specific values that are valid with respect to its underlying probability distribution. We can use random variables wherever a normal deterministic variable might be used, but if we add a random variable with a deterministic variable, we will get a new random variable.

For example, if we think tomorrow's temperature is just going to be today's temperature plus some random noise, we can model this as $T = t_0 + e$, where *e* is a random variable of noise. The noise might have *normal (Gaussian) distribution* centered around 0 with a variance of 1. Thus *T* will be a new normal distribution with mean t_0 (today's temperature), but it will still have a variance of 1. A normal distribution is the familiar bell-shaped curve.

Table 7.3 shows a few common distributions. The normal distribution gets wider or narrower depending on the variance parameter, but otherwise it looks the same for any set of parameters. In contrast, the beta and gamma distributions can look quite different depending on their parameters—two different versions of each of these are shown.

Table 7.3. Common probability distributions